

Time: 03 Hours



Marks: 80

- Note: 1. Question 1 is compulsory
 2. Answer any three out of the remaining five questions.
 3. Assume any suitable data wherever required and justify the same.

- Q1 a) Mention four characteristics of big data and explain in detail. [5]
 b) Explain Shuffle & Sort phase and Reducer phase in Map Reduce. [5]
 c) Demonstrate how business problems have been successfully solved faster, cheaper and more effectively considering NoSQL Google's Bigtable case study. Also illustrate the business drivers and the findings in it. [5]
 d) List down all six constraints that must be satisfied for representing a stream by buckets using DGIM algorithm with examples. [5]

- Q2 a) The project manager at XYZ Ltd., Ms. Meera, is responsible for maintaining details of all active projects. She has organized the project information in the following table: [10]

Project Id	Project Name	Budget	Status
1	CRM Implementation	120000	In Progress
2	Cloud Infrastructure	180000	Completed
3	Network Upgrade	60000	Not Started
4	E-Commerce Platform	220000	Completed
5	Data Analytics	90000	In Progress

- i) Create a Data frame in R for the above project data and display the output.
 ii) Ms. Meera has recently approved 2 new projects and wants to add their information. The new projects are as follows:

Project Id	Project Name	Budget	Status
6	UX Research	160000	Not Started
7	Cloud Integration	190000	Not Started

Update the Data frame to include the new projects and demonstrate the final output.

- b) Write a short note on variations of NoSQL architectural patterns. [10]

Q.P. Code:-

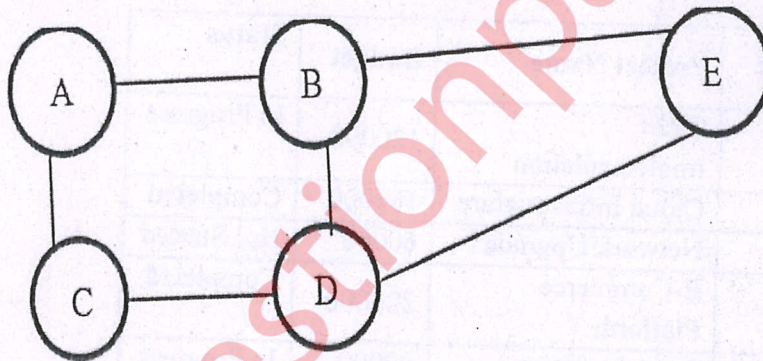
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Program Code:-

1T00737

- Q3 a) Suppose the stream is $S = \{10, 12, 8, 15, 6, 9, 14, 7\}$. Let hash functions $h(x) = 5x + 11 \bmod 32$ for some a and b , treat result as a 5-bit binary integer. Show how the Flajolet- Martin algorithm will estimate the number of distinct elements in this stream. [10]
- b) Explain natural join and grouping and aggregation relational algebraic operation using MapReduce. [10]
- Q4 a) Write a map reduce pseudo code to solve the word count problem. Apply map reduce working on the following document: "Big data is powerful. Big data drives decisions." [10]
- b) With a neat sketch, explain the architecture of the data-stream management system. [10]
- Q5 a) Determine communities for the given social network graph using Girvan- Newman algorithm. [10]



- Q6 a) Describe the components of Hadoop ecosystem with the help of a diagram. [10]
b) What is recommendation system? How is classification algorithm used in recommendation system? [10]



BE/comp/sem-VII/CBCGS/R-19/c-scheme/BDA/SH-24

Time: 3 Hours

Total Marks: 80

Note: 1. Question 1 is compulsory

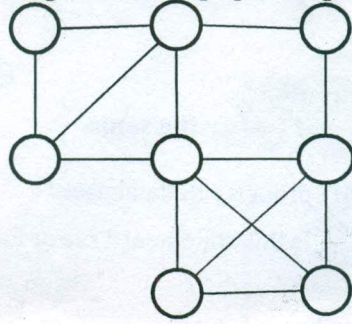
2. Answer any three out of the remaining five questions.

3. Assume any suitable data wherever required and justify the same.

OP-10068/56

- Q.1 a) Explain CAP. How is CAP different from ACID property in databases? [05]
- ✓ b) Secondary Name node is a backup of Name node. Is this statement True or False? Justify your answer. [05]
- ✓ c) List and explain the core business drivers behind the NoSQL movement. [05]
- d) List down any five constraints that must be satisfied for representing a stream by buckets using DGIM algorithm with examples. [05]
- Q.2 a) List the architectural patterns in NoSQL databases. Discuss the Key-Value and Document-Oriented patterns, focusing on their characteristics, use cases, and examples. [10]
- b) Write a map reduce pseudo code for word count problem. Apply map reduce working on the following document: [10]
- "This is NoSQL. NoSQL handles complex data."
- Q.3 a) Explain Map Reduce execution pipeline with suitable example. [10]
- b) Create a Bloom filter with the following parameters: [10]
- Size of the bit array $m=8$
Hash functions:
- $h_1(x)=x \bmod m$
 $h_2(x)=(2x+1) \bmod m$
 $h_3(x)=(3x+2) \bmod m$
- (i) Insert the following elements into the Bloom filter: 12, 25, 30, 5
(ii) Check if the following elements are present in the Bloom filter: 6, 55
(iii) Discuss the results of your checks, identifying which elements is true positive and which is true negative.
- Q.4 a) For the stream of integers: 9, 8, 7, 6, 5, 4, 3, 2. Use the hash function, $h(x)=(2x+1) \bmod 32$ and treat the result as a 5-bit binary integer. Show the steps of the Flajolet-Martin algorithm to estimate the number of distinct elements in this stream. [10]
- b) Draw a diagram of the typical Hadoop Ecosystem and explain any two components of it. [10]

- Q.5 a) Write an algorithm for the Clique Percolation Method and discover the communities in the given below graph using Clique Percolation Method with clique $k=3$. [10]



- b) i. List and explain the functions provided by R to combine different sets of data. [10]
 ii. Write the script to sort the values contained in the following vector in ascending order and descending order: (46, 23, 15, 38, 98, 56, 28, 78). Demonstrate the output.

- Q.6 a) The project manager at ABC Corp, Mr. Thomas, needs to track information about ongoing projects in the organization. He has the following details about current projects in a table format: [10]

ProjectId	ProjectName	Budget
1	Website Redesign	150000
2	Mobile App Launch	100000
3	Data Migration	80000
4	AI Development	200000
5	Cybersecurity Audit	50000

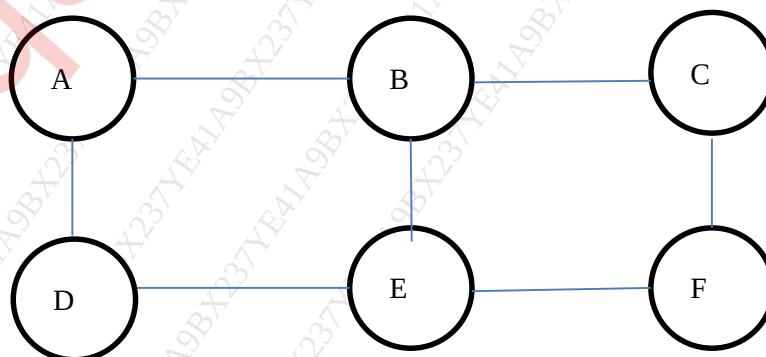
- i) Create a Data frame in R for the above project data and display the output.
 ii) Show the structure and summary statistics of the Data Frame created.
 b) Justify the use of a Content-Based Recommendation System with a specific case study. [10]

Time: 03 Hours

Marks: 80

Note: 1. Question 1 is compulsory**2. Answer any three out of the remaining five questions.****3. Assume any suitable data wherever required and justify the same.**

- Q1 a) Explain how big data problems are handled by Hadoop system. [5]
 b) Mention four characteristics of big data and explain in detail. [5]
 c) List and explain the core business drivers behind the NoSQL movement. [5]
 d) Explain the concept of bloom filter with an example. [5]
- Q2 a) What is graph store? Give an example where a graph store can be used to effectively solve a particular business problem. [10]
 b) Write a map reduce pseudo code for word count problem. Illustrate with an example showing all the steps. [10]
- Q3 a) Suppose the stream is $S = \{4, 2, 5, 9, 1, 6, 3, 7\}$. Let hash functions $h(x) = 3x + 7 \bmod 32$ for some a and b , treat result as a 5-bit binary integer. Show how the Flajolet- Martin algorithm will estimate the number of distinct elements in this stream. [10]
 b) Describe applications of data visualization. [10]
- Q4 a) Explain selection and projection relational algebraic operation using MapReduce. [10]
 b) Explain DGIM algorithm for counting ones in a stream with example. [10]
- Q5 a) Determine communities for the given social network graph using Girvan- Newman algorithm. [10]



- b) Consider the following data frame given below:

[10]

course	id	class	marks
1	11	1	56
2	12	2	75
3	13	1	48
4	14	2	69
5	15	1	84
6	16	2	53

- i. Create a subset of course less than 5 by using [] brackets and demonstrate the output.
- ii. Create a subset where the course column is less than 4 or the class equals to 1 by using subset () function and demonstrate the output.

- Q6 a) i. Write a script to create a dataset named data1 in R containing the following text. [10]

Text: 2, 3, 4, 5, 6.7, 7, 8.1, 9

- ii. Explain the various functions provided by R to combine different sets of data.

- b) Describe collaborative filtering in recommendation system.

[10]

Time: 3 Hours

Marks: 80

Note: 1. Question 1 is compulsory

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3. Assume any suitable data wherever required and justify the same.

- Q1**
- a) Distinguish between Name node and Data node. [5]
 - b) List and explain the core business drivers behind the NoSQL movement. [5]
 - c) Mention four characteristics of big data. Elaborate these characteristics with respect to social media websites. [5]
 - d) List and explain the different issues and challenges in data stream query processing. [5]

- Q2**
- a) What is a key-value store? What are the benefits of using a key-value store? [10]
 - b) Write a map reduce pseudo code to multiply two matrices. Apply map reduce working to perform following matrix multiplication. [10]

$$\begin{array}{cccc} 1 & 2 & 6 & 7 \\ & & X & \\ 3 & 4 & 8 & 9 \end{array}$$

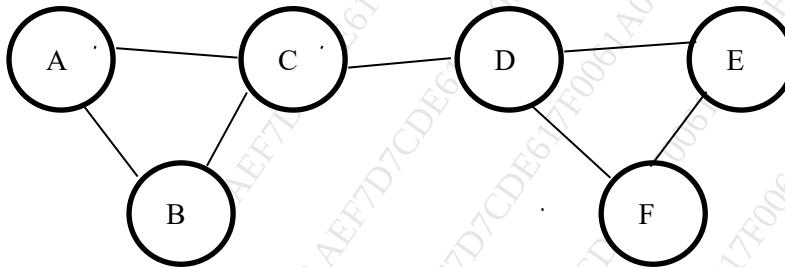
- Q3**
- a) Suppose the stream is $S = \{2, 1, 6, 1, 5, 9, 2, 3, 5\}$. Let hash functions $h(x) = ax + b \bmod 16$ for some a and b , treat result as a 4-bit binary integer. Show how the Flajolet- Martin algorithm will estimate the number of distinct elements, $h(x) = 4x + 1 \bmod 16$. [10]

- b) Consider the following data frame given below: [10]

course	id	class	marks
1	11	1	56
2	12	2	75
3	13	1	48
4	14	2	69
5	15	1	84
6	16	2	53

- i. Create a subset of course less than 3 by using [] brackets and demonstrate the output.
 - ii. Create a subset where the course column is less than 3 or the class equals to 2 by using subset () function and demonstrate the output.
- Q4**
- a) Explain natural join and grouping and aggregation relational algebraic operation using MapReduce. [10]
 - b) With a neat sketch, explain the architecture of the data-stream management system. [10]

- Q5 a)** Determine communities for the given social network graph using Girvan-Newman algorithm. [10]



- b) List and discuss various types of data structures in R. [10]

- Q6 a)** i. The following table shows the number of units of different products sold on different days: [10]

Product	Monday	Tuesday	Wednesday	Thursday	Friday
Bread	12	3	5	11	9
Milk	21	27	18	20	15
Cola Cans	10	1	33	6	12
Chocolate bars	6	7	4	13	12
Detergent	5	8	12	20	23

Create five sample numeric vectors from this data.

- ii. Name and explain the operators used to form data subsets in R.
- b) Define collaborative filtering. Using an example of an e-commerce site like flipkart or amazon describe how it can be used to provide recommendation to users. [10]



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Note: 1. Question 1 is compulsory

2. Answer any three out of the remaining five questions.
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Q1 ✓ a) What is the basic difference between traditional RDBMS and Hadoop? [5]

✓ b) What are the 3 V's of big data? Give two big data case studies indicating respective V's with justification. [5]

✓ c) Explain how node failure is handled in Hadoop. [5]

✓ d) List down all six constraints that must be satisfied for representing a stream by buckets using DGIM algorithm with examples. [5]

Q2 ✓ a) Describe the four ways by which big data problems are handled by NoSQL. [10]

✓ b) Write a map reduce pseudo code to multiply two matrices. Apply map reduce working to perform following matrix multiplication. [10]

$$M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \quad X \quad V = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Q3 ✓ a) Suppose the stream is $S = \{4, 2, 5, 9, 1, 6, 3, 7\}$. Let hash functions $h(x) = x + 6 \pmod{32}$ for some a and b , treat result as a 5-bit binary integer. Show how the Flajolet- Martin algorithm will estimate the number of distinct elements in this stream. [10]

✓ b) i. Create a data frame from the following 4 vectors and demonstrate the output: [10]

```
emp_id = c(1:5)
emp_name = c("Rick", "Dan", "Michelle", "Ryan", "Gary")
start_date = c("2012-01-01", "2013-09-23", "2014-11-15", "2014-05-11", "2015-03-27")
salary = c(60000, 45000, 75000, 84000, 20000)
```

✓ ii. Display structure and summary of the above data frame.

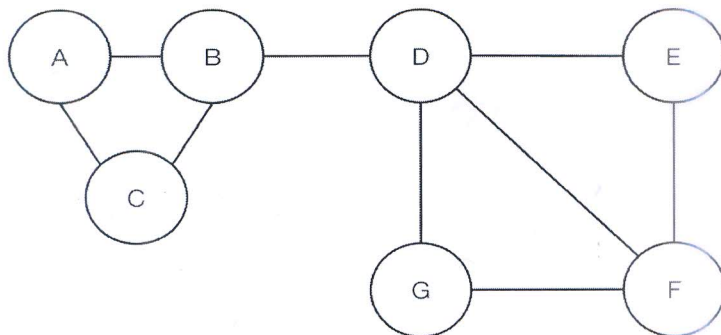
✓ iii. Extract the emp_name and salary columns from the above data frame.

✓ iv. Extract the employee details whose salary is less than or equal to 60000.

Q4 ✓ a) Explain Map Reduce execution pipeline with suitable example [10]

✓ b) Explain DGIM algorithm for counting ones in a stream with example. [10]

- Q5 a) Determine communities for the given social network graph using Girvan-Newman algorithm. [10]



- ✓ b) List and explain various functions that allow users to handle data in R workspace with appropriate examples. [10]

- Q6 a) i. What are the advantages of using functions over scripts? [10]

- ✓ ii. Suppose you have two datasets A and B.
Dataset A has the following data: 6 7 8 9.
Dataset B has the following data: 1 2 4 5.
Which function is used to combine the data from both datasets into dataset C.
Demonstrate the function with the input values and write the output.

- ✓ b) How recommendation is done based on properties of the product? Explain with the help of an example. [10]

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Note: 1. Question 1 is compulsory

2. Answer any three out of the remaining five questions.

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Q1 a) What is function of Map Tasks in the Map Reduce framework? Explain with the help of an example. [5]

b) Demonstrate how business problems have been successfully solved faster, cheaper and more effectively considering NoSQL Google's MapReduce case study. Also illustrate the business drivers and the findings in it. [5]

c) Why is HDFS more suited for applications having large datasets and not when there are small files? Elaborate. [5]

d) Explain the concept of bloom filter with an example [5]

Q2 a) Name the three ways that resources can be shared between computer systems. Name the architecture used in big data solutions and describe it in detail. [10]

b) Write a map reduce pseudo code for word count problem. Apply map reduce working on the following document: [10]

"This is an apple. Apple is red in color".

Q3 a) Suppose the stream is 1, 3, 2, 1, 2, 3, 4, 3, 1, 2, 3, 1. Let $h(x) = 6x + 1 \pmod{5}$. Show how the Flajolet- Martin algorithm will estimate the number of distinct elements in this stream. [10]

b) Consider the following data frame given below: [10]

subject	class	marks
1	1	56
2	2	75
3	1	48
4	2	69
5	1	84
6	2	53

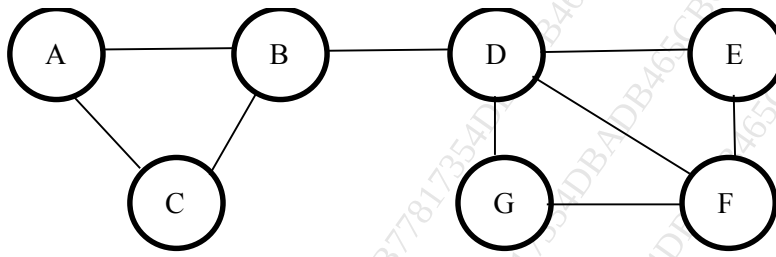
i. Create a subset of subject less than 4 by using subset () function and demonstrate the output.

ii. Create a subset where the subject column is less than 3 and the class equals to 2 by using [] brackets and demonstrate the output.

Q4 a) What are the Core Hadoop components? Explain in detail. [10]

b) With a neat sketch, explain the architecture of the data-stream management system. [10]

Q5 a) Determine communities for the given social network graph using Girvan- Newman algorithm. [10]



- b) The data analyst of Argon technology Mr. John needs to enter the salaries of 10 employees in R. The salaries of the employees are given in the following table: [10]

Sr. No.	Name of employees	Salaries
1	Vivek	21000
2	Karan	55000
3	James	67000
4	Soham	50000
5	Renu	54000
6	Farah	40000
7	Hetal	30000
8	Mary	70000
9	Ganesh	20000
10	Krish	15000

- Which R command will Mr. John use to enter these values demonstrate the output.
- Now Mr. John wants to add the salaries of 5 new employees in the existing table, which command he will use to join datasets with new values in R. Demonstrate the output.

- Q6 a) i. Write the script to sort the values contained in the following vector in ascending order and descending order: (23, 45, 10, 34, 89, 20, 67, 99). Demonstrate the output. [10]
- ii. Name and explain the operators used to form data subsets in R.
- b) How recommendation is done based on properties of product? Elaborate with a suitable example. [10]
